

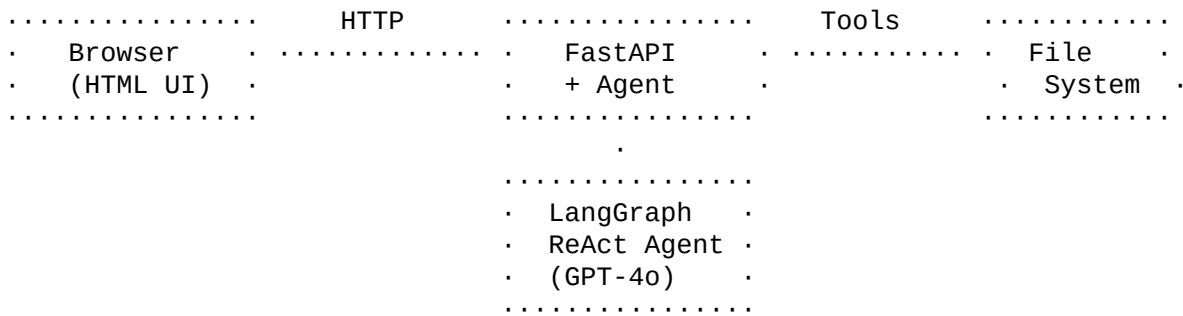
Build And Deploy AI Agent

A **note-taking AI agent** built with **LangGraph** and exposed via a **FastAPI** web interface with HTML templates · demonstrates building and deploying a functional AI agent.

· Features

- **·ReAct Agent** · LangGraph-based reasoning + acting agent
 - **·File I/O Tools** · Read and write notes through natural language
 - **·Web Interface** · FastAPI + Jinja2 HTML templates
 - **·GPT-4o-mini** · Powered by OpenAI efficient model
-

.. Architecture



· File Structure

```
BuildAndDeployAIAgent/
... main.py           # FastAPI server with routes and Jinja2
templates
... agent.py          # LangGraph ReAct agent with read/write
tools
... requirements.txt  # Dependencies
... note.txt          # Persistent note storage
... templates/        # HTML templates for web UI
... public/           # Static assets (CSS, JS)
```

· Getting Started

Prerequisites

- Python 3.10+
- OpenAI API key

Installation

```
cd BuildAndDeployAIAgent
pip install -r requirements.txt
```

Environment Variables

Create a `.env` file:

```
OPENAI_API_KEY=sk-...
```

Run

```
python main.py
```

Visit `http://localhost:8000` in your browser.

· Logic Flow

- 1 **User Input** · User types a natural language request in the web UI
- 2 **API Route** · FastAPI receives the request as an `AgentRequest`
- 3 **Agent Processing** · LangGraph ReAct agent interprets the request
- 4 **Tool Selection** · Agent decides which tool to use:
 - `write_note(note)` · Write content to `note.txt`
 - `read_note()` · Read current content of `note.txt`
- 5 **Tool Execution** · Selected tool performs file I/O
- 6 **Response** · Agent formulates response based on tool output
- 7 **Display** · Response rendered in the HTML UI

Key Components

Component	File	Purpose
<code>AgentRequest</code>	<code>main.py</code>	Pydantic model for incoming requests
<code>AgentResponse</code>	<code>main.py</code>	Pydantic model for responses
<code>write_note()</code>	<code>agent.py</code>	Tool to write notes to file
<code>read_note()</code>	<code>agent.py</code>	Tool to read notes from file
<code>create_react_agent()</code>	<code>agent.py</code>	LangGraph agent factory

· Dependencies

```
fastapi
uvicorn
jinja2
langchain
langchain-openai
langgraph
python-dotenv
```

· License

Educational project · use freely for learning and reference.

